



Society of Petroleum Engineers

SPE-227671-MS

Assessing Geomechanical Risks in Large-Scale Reservoirs: A Two-Way Coupled Geomechanics Modeling Approach with Integrated Safety Factor Screening for CO₂ Storage

Xiaolin. Chen and Abdulaziz M. Baiz, Saudi Aramco, Dhahran, Eastern Province, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227671-MS](https://doi.org/10.2118/227671-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper addresses the geomechanical risks associated with CO₂ sequestration in large-scale, highly heterogeneous reservoirs. A novel 2-Way Coupled Geomechanics Modeling approach is developed to estimate the effect of fault reactivation caused by CO₂ injection on the pressure field in nearby rock formations. This approach integrates a Safety Factor (SF) to evaluate the hydro-mechanical stability of adjacent faults under different injection rates.

The proposed methodology utilizes a 2-way coupled geomechanics and reservoir simulator to evaluate reservoir behavior for both the concept model SPE1 and a large complex synthetic model, incorporating a robust discontinuity treatment to accurately capture the effects of faults and fractures on reservoir performance. This approach enables the simulation of large-scale models, providing a comprehensive understanding of the reservoir's behavior during CO₂ injection. A safety factor-based screen tool is employed to rapidly assess formation integrity, facilitating swift risk identification and evaluation of the interaction between fault and reservoir permeabilities.

The results highlight the significant impact of reservoir heterogeneity on the assessment of reservoir integrity during CO₂ injection. A safety factor-based approach is used to evaluate the integrity of both the SPE1 concept model and the large synthetic sector model, considering both cohesion and cohesionless faults. The study demonstrates that when the safety factor of all grid cells remains greater than 1, the reservoir integrity is preserved, and provides insight into how geological configuration, geomechanical properties, and operating conditions influence the risk of fault reactivation.

The integration of advanced numerical techniques with the safety factor-based screen tool enables rapid identification of formation failure and CO₂ leakage risks, while sensitivity analyses provide a robust framework for future studies on fault stability and induced seismicity. This comprehensive approach ultimately aids in the development of more effective risk assessment and mitigation strategies, ensuring the safe and efficient storage of CO₂ in heterogeneous reservoirs.

Introduction

The injection of large volumes of CO₂ into deep geological formations is a critical component of efforts to reduce greenhouse gas emissions and mitigate climate change, offering a promising solution for long-term carbon sequestration. Among various CO₂ storage options as shown in Fig. 1 (Ali, M., et al. 2022), saline aquifers emerge as one of the most promising methods due to their advantageous geological properties and widespread global distribution. Saline aquifers are composed primarily of porous sandstone and carbonate rocks, which provide high porosity and permeability, making them ideal for CO₂ storage. The porous nature of these rocks allows CO₂ to be stored in the interstitial spaces between rock grains, significantly enhancing storage capacity. Typically located at depths between 800 to 3000 meters, these aquifers ensure that CO₂ remains in a supercritical state, which is denser than gas but lighter than liquid, thereby optimizing storage capacity and minimizing escape risks. Given these advantages, our study has opted to develop a large synthetic model for CO₂ injection into a saline aquifer to deepen understanding of fault reactivation and related geological changes, proposing effective management strategies for stable CO₂ storage.

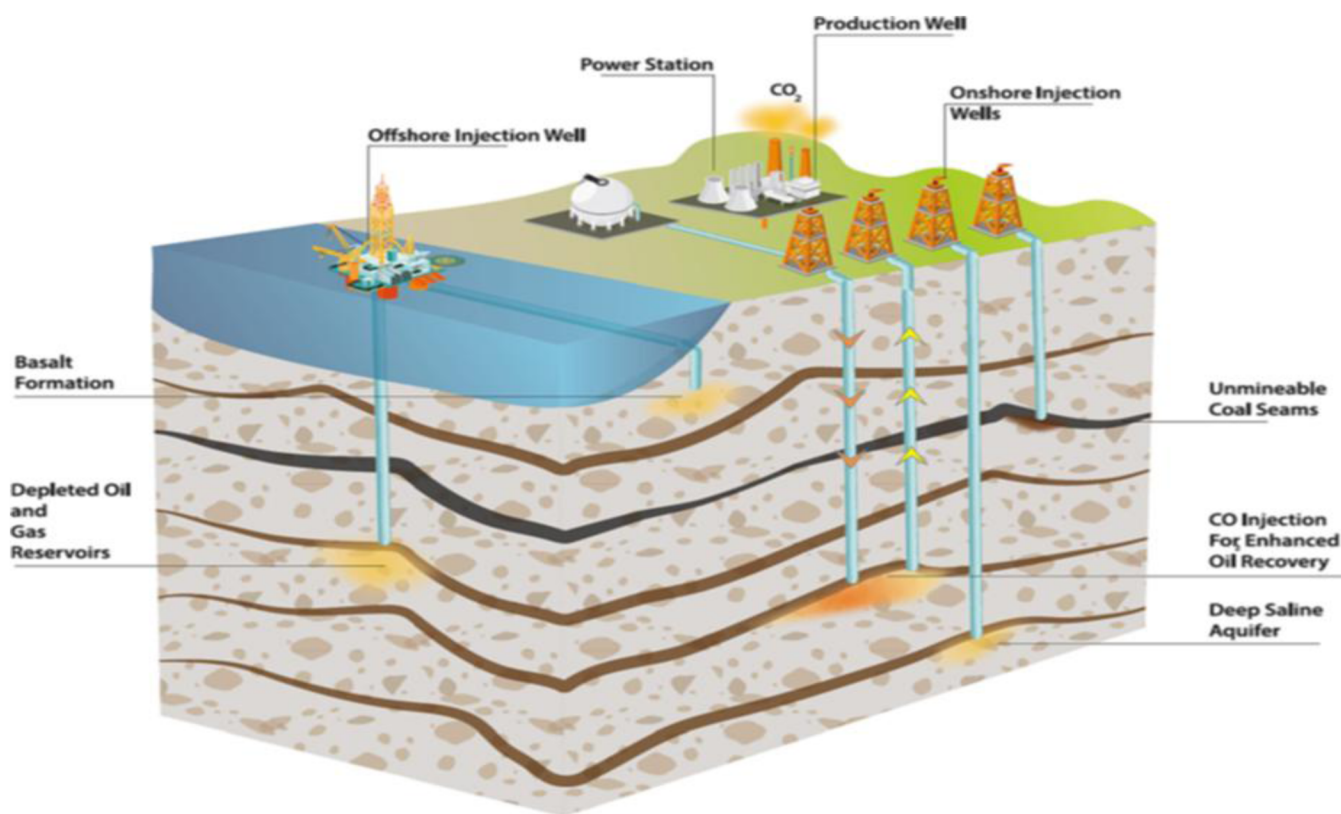


Figure 1—CO₂ storage options (Ali, M., et al. 2022)

However, one of the primary challenges in CO₂ sequestration is maintaining reservoir integrity to avoid fault reactivation and ensure the containment of CO₂ within the target storage zone. Fault reactivation is a particular concern, as changes in the stress state caused by CO₂ injection can reactivate dormant faults, creating pathways for CO₂ to escape, especially in reservoirs with pre-existing fractures or faults. Reservoir deformation can pose a significant risk to the integrity of the cap rock, which serves as a critical seal to prevent CO₂ migration. Ensuring the long-term success of CO₂ storage requires a comprehensive understanding of these geomechanical risks and their potential to impact the integrity of the reservoir. Several approaches have been proposed in the literature (Kanin et al. 2023, Gillioz et al. 2024, Ferrari et al. 2024, Shi et al. 2024) to address these challenges, including the use of coupled flow and geomechanical

models, fault stability analysis, and risk assessment frameworks. However, these methods often face limitations in terms of computational efficiency and their ability to handle large-scale numerical models.

To accurately model the complex interactions between fluid flow and rock deformation in large-scale reservoirs, our in-house simulator leverages cutting-edge numerical methods and high-performance computing capabilities, enabling the efficient simulation of massive reservoir models with intricate geological features. In response, we have developed an integrated approach that combines two-way coupling of reservoir flow and geomechanics simulations. The data exchange between the two in-house simulators is performed using the multiple program multiple data (MPMD) paradigm, allowing for a more accurate prediction of the interactions between CO₂ injection and reservoir deformation. To simplify the complex geological settings, an equivalent continuous material approach was applied to model faults and discrete fractures by integrating fractures, faults, and the surrounding matrix into a continuum. Additionally, we utilize safety factors based on the Mohr-Coulomb failure criterion to assess geomechanical responses and evaluate fault activation risks.

The capability of our approach is demonstrated through two case studies. We first consider a simplified SPE1 case to demonstrate the capability of handling reservoir grids with faults, investigating the difference between 1-way and 2-way coupling, and fault instability during CO₂ injection. Later, we consider CO₂ injection into an extra-large multi-fault reservoir model to assess geomechanical responses and evaluate fault activation risks using safety factors. This case study demonstrates the key advantage of our ability to handle large-scale numerical models, with up to 15 million cells, making it suitable for field-scale applications. By integrating advanced computational techniques with geomechanical and reservoir engineering principles, our methodology provides a robust framework for assessing and mitigating the risks associated with CO₂ storage in saline reservoirs.

Methodology

Two-way coupled fluid flow and geomechanics simulation

The integrated geo-mechanical workflow employed in this study consists of the following steps. First, a 3D structural geomodel is constructed to provide a detailed representation of the reservoir's geological structure. Next, a dynamic reservoir model and a geo-mechanical model are built to simulate the behavior of the reservoir during CO₂ injection. The in-house reservoir simulator is then used to simulate fluid injection and the resulting pressure changes within the reservoir over multiple time steps. The updated stress fields obtained from the reservoir simulation are subsequently used as inputs for the in-house geomechanics simulator, which calculates the updated stresses, strains, and displacements within the reservoir. Finally, a parametric study is conducted to assess the fault stability using safety factors, as well as surface uplift or subsidence, in order to evaluate the reservoir's integrity during CO₂ injection.

Coupling between fluid flow and geomechanics. Coupled hydro-mechanical analysis can be realized through two types of solutions: one-way and two-way coupling. In one-way coupling, the reservoir model is simulated for a given timeframe to generate a series of pressure distributions, which are then supplied to a geomechanical model allowing for the update of the subsurface stress field. This process is straightforward to implement and enables the use of preferred software for reservoir simulation and geomechanics simulation, respectively. However, the communication is only one-way, from fluid flow to mechanics, where pressure and saturation data are sent from the reservoir simulator to the geomechanical module, but no information is fed back to the flow simulator. As a result, the changes in stresses and strains do not impact the values of fluid pressure calculated by the reservoir simulator, making one-way coupling poorly accurate for modeling the effect of deformation on the petrophysical properties of the porous medium. This method can only be used to assess the general effects and magnitude of stresses, while cannot capture the complex interactions between fluid flow and geomechanics.

Two-way coupling can be categorized into two approaches: full coupling, also known as the fully implicit method, and partial coupling, also referred to as sequential iterative coupling. The full coupling approach involves solving the flow and stress equations simultaneously using the Newton-Raphson method, which provides an accurate solution but is computationally expensive due to the requirement for a unified flow-geomechanics simulation framework. In contrast, sequential iterative coupling methods involve a more iterative process, where the fluid flow equations are solved and the results are sent to the geomechanical simulator to compute stresses, strains, and displacements. The updated porosity and permeability are then fed back to the flow simulator, and this process is repeated until convergence is achieved, at which point the simulation proceeds to the next timestep. This sequential iterative coupling method offers a more efficient and flexible alternative to full coupling, making it a suitable choice for large-scale reservoir simulations.

Implementation of sequential coupling. Building on our previous work (L. Qiao et al, 2024), which provides a detailed description of the governing equations and numerical methods employed in our in-house geomechanics simulator, this paper briefly explains the two-way sequential coupling using the Multiple Program Multiple Data (MPMD) paradigm to facilitate reader understanding, with a focus on its application to CO₂ injection case study.

The MPMD program enables simultaneous loading of the flow simulator and geomechanics simulator, facilitating data exchange between them through message passing interface (MPI) function calls. The MPMD communication pattern is established at the beginning of the simulation and remains static throughout the run. To facilitate communication, the geomechanics simulator creates two MPI partitions: one for reservoir grids, which interacts with the flow simulator, and another for geomechanics grids. The solution array is exchanged directly between the simulators using MPMD.

An example of the two-way coupling workflow using MPMD is illustrated in Fig. 2. In this example, a total of 9 cores are utilized, with 3 cores dedicated to the flow simulator and 6 cores allocated to the geomechanics simulator. Within the geomechanics simulator, 4 cores are assigned to communicate with the flow simulator, while the remaining 2 cores are used for a partition containing only burden cells. This configuration highlights MPMD's ability to enable seamless communication between the flow simulator and the geomechanics simulator, significantly improving performance compared to the traditional approach of using restart file-based data exchange between the two simulators.

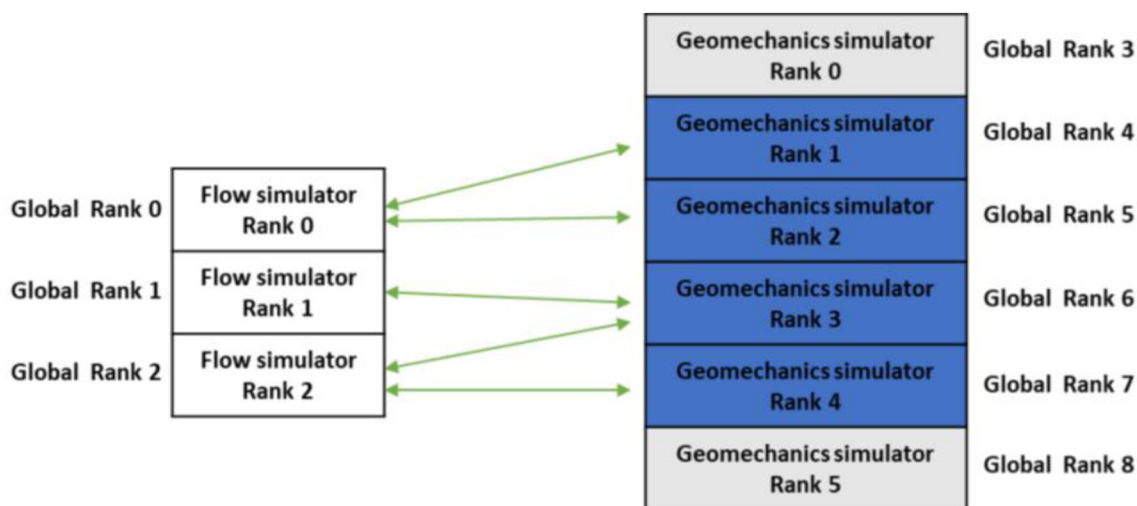


Figure 2—A typical MPMD communication pattern (L. Qiao et al, 2024)

The system of equations for fluid flow and geomechanical changes is solved sequentially, with the flow problem addressed first, followed by the mechanical problem. At each coupling step, the reservoir simulator computes pressure changes from CO₂ injection and passes them to the geomechanics simulator

to assess stress, strain, and deformation. The geomechanics simulator then uses a momentum equation to determine changes in porosity (ϕ) and permeability (k) due to volumetric deformations. These updated values, specifically the permeability multiplier defined with respect to initial permeability, are sent back to the fluid flow simulator, where they are applied at the beginning of the next time step to update transmissibility and well productivity index (PI) accordingly.

Faults modeling and stability evaluation

The Mohr-Coulomb failure criteria has been widely and successfully used for assessing the integrity of the targeted reservoir during CO₂ injection, as demonstrated in Fig. 3. As fluid is injected into the subsurface, where pre-existing faults are present, the pore pressure (P) increases, thereby reducing the effective normal stress acting on the fault. This reduction causes the Mohr circle to shift to the left, moving closer to the failure line, which indicates an increased likelihood of fault reactivation.

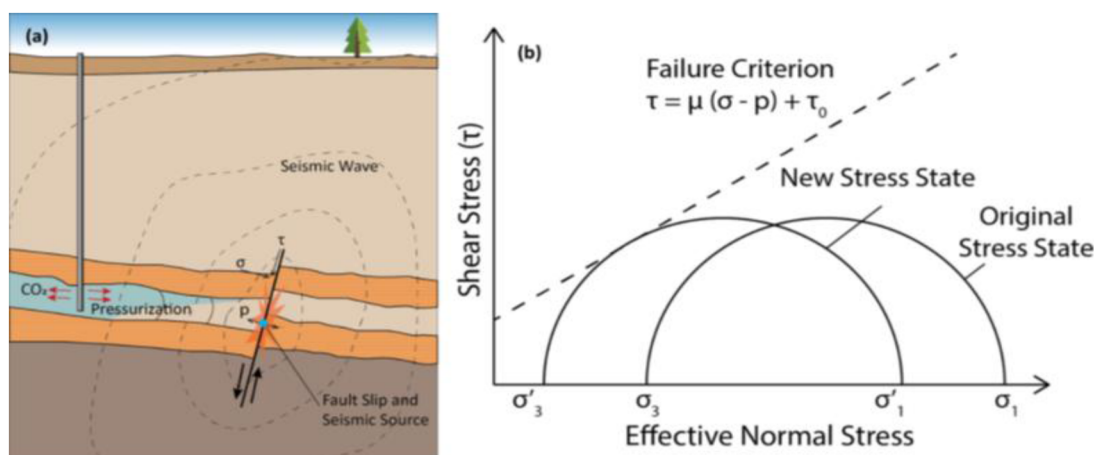


Figure 3—Induced fault slip caused by pore pressure and stress changes with CO₂ injection (left), and Mohr diagram representation of the fault failure criterion (right) (T. Xiao. et al, 2024)

To evaluate fault slip and reactivation, three primary fault modeling methods are available: analytical, semi-analytical, and numerical approaches (Zoback, 2007). The analytical approach simplifies the problem by assuming homogeneous reservoir geometry and spatially constant reservoir properties. While this approach provides valuable insights, it is limited in its ability to account for real-world complexity. The semi-analytical approach addresses some of these limitations by incorporating spatially variable reservoir properties, though it still relies on many of the assumptions of the analytical method. Numerical methods, such as the Finite Element Method (FEM), provide significant advantages, particularly for evaluating irregularly shaped reservoirs with heterogeneous properties—a key consideration for this research, given the complex geological structures and heterogeneous characteristics of oil and gas reservoirs. The proposed 3D numerical modeling approach effectively addresses discontinuities, providing fault displacements and strains, and enhances the accuracy and reliability of fault modeling and stability evaluations by integrating with geological and reservoir simulations.

An equivalent continuum model for fractures and faults (fault-treatment). The discontinuity in the reservoir grid need to be treated before the finite element method can be applied. This work implements a 3D continuum representation of fault zones, modeling faults within elements (cells) by integrating the fault core and surrounding matrix rock. Discrete fractures share the same physical properties as faults and utilize the same modeling procedure. The permeability update for stress-sensitive faults in two-way coupling is also discussed, due to its significant impact on the flow field.

The properties for elements with faults are represented using the ‘equivalent material’ constitutive relation (G.N. Pande et al.), which combines the stiffness of intact rock and faults as:

$$C = \left[(C^m)^{-1} + \sum_l (C_l^f)^{-1} \right]^{-1} \quad (1)$$

In this model, faults are assumed to be planar. The stiffness matrix for intact rock is denoted as C^m , and C_l^f represents the stiffness matrix for the l -th fault. The fault stiffness matrix C^f is determined by the fault plane's normal stiffness (kn), shear stiffness (ks), dip angles ($\theta 1$, $\theta 2$), and the spacing of fault planes. Here, $\theta 1$ is the polar angle, and $\theta 2$ is the azimuthal angle.

During the permeability update for the 3D model, the initial permeability is separated into intact rock and faults components, which are then updated separately.

$$K_0 = K_0^m + K_0^f = \begin{bmatrix} K_{0x}^m & 0 & 0 \\ 0 & K_{0y}^m & 0 \\ 0 & 0 & K_{0z}^m \end{bmatrix} + \sum_l \begin{bmatrix} K_{0xl}^f & 0 & 0 \\ 0 & K_{0yl}^f & 0 \\ 0 & 0 & K_{0zl}^f \end{bmatrix} \quad (2)$$

where K_0 is the initial permeability tensor, with K_0^m representing intact rock and K_0^f representing the faults. The permeability components for intact rock are K_{0x}^m , K_{0y}^m and K_{0z}^m in the x, y, and z directions, respectively. For the l -th faults, the components are K_{0xl}^f , K_{0yl}^f and K_{0zl}^f in the x, y, and z directions, respectively. The off-diagonal contributions are not considered in this model.

The permeability update is determined separately for intact rock and faults, each employing distinct correlations. For intact rock, the Kozeny-Carman correlation is utilized, defining permeability as a function of porosity. For faults, the permeability change is determined based on the fault's normal and shear strain. Readers interested in detailed information on fracture permeability correlations are referred to the (VISAGE Technical Description) for further reading.

An example of fault modeling is illustrated in Fig. 4. The reservoir grid, refined to an 18x18x3 grid based on the SPE1 grid, features two faults: one oriented along the X-axis and the other along the Y-axis, intersecting in the middle of the reservoir. The fault parameters include a normal stiffness of 1.7683e5 psi/ft and a shear stiffness of 0.66e5 psi/ft. The dip angle pairs are (90°, 90°) for the X-axis fault and (90°, 0°) for the Y-axis fault. After treating the discontinuity in the reservoir grid, the resulting smooth grid is shown in Fig. 4 (right). Our in-house geomechanics simulator supports overburden and under-burden extensions. However, these extensions are not included in this study.

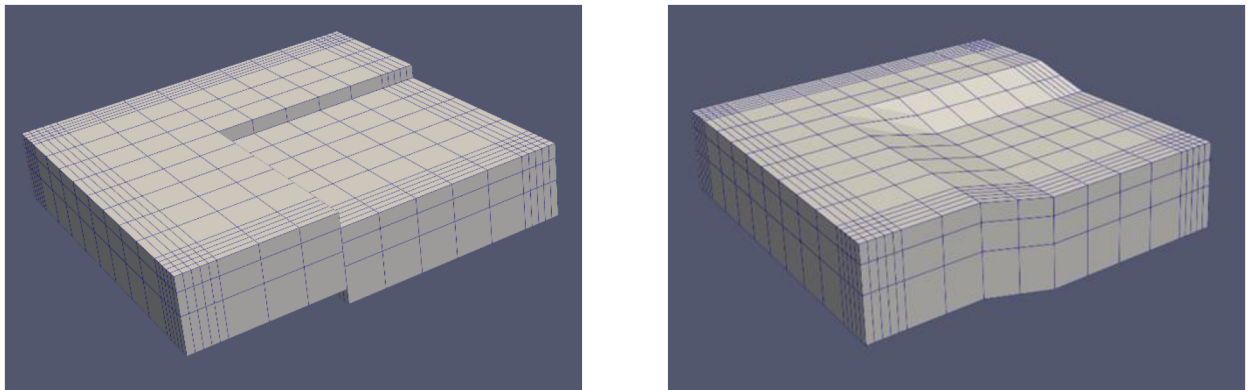


Figure 4—Grids before the discontinuity treatment (left) and after the discontinuity treatment (right)

The horizontal and vertical stress fields, along with the faults' normal and shear displacements, simulated by our in-house geomechanics simulator, were validated against a commercial software in our previous work (L. Qiao et al., 2024). These results confirm the consistency between the two tools in predicting fault

displacements, which are crucial for determining permeability changes in two-way coupling. It is important to note that permeability updates for fault effects can change notably during production in two-way coupled simulations. When fault opening increases, permeability along the fault plane increases markedly. After reaching a certain threshold, the process stabilizes, and sliding displacement becomes dominant. Fault sliding can break asperities, creating debris that blocks the flow path and consequently reduces permeability. In this study, the fault normal and shear strain threshold is set to 0.0005, with a permeability increasing factor of 100 and a reducing factor of 0.01. The detail of permeability update will be discussed later in the numerical model section.

Risk assessment through safety factor. When assessing the stability of CO₂ geological sequestration sites, faults are typically modeled as ideal elastic-plastic bodies. The Mohr-Coulomb criterion is particularly useful for assessing slip tendency, as it describes shear failure in soil and rock by providing a linear relationship between shear strength and effective normal stress acting on the fault. This approach offers two key advantages: it requires fewer computational parameters, as the onset of fault zone reactivation is controlled primarily by rock cohesion and internal friction angle, and it allows for straightforward application of evaluation indicators to determine fault stability. For intact rock in this study, the cohesion value is set to 435 psi. Since faults are typically viewed as pre-existing natural discontinuities, their cohesion is considered negligible. The friction angle usually ranges from 30° to 39°.

We analyze fault stability by examining the variation in the safety factor, defined as follows. The yield surface, which defines the boundary between stable and unstable conditions, can be expressed as:

$$\tau_n = \sigma_n^{eff} \tan \Phi + c \quad (3)$$

Where ϕ is the friction angle and c is cohesion.

To analyze fault stability, we utilize the safety factor (SF), defined as the ratio of the distance from the center of Mohr's circle to the yield surface $|AF|$ to the radius of Mohr's circle $|AE|$ as shown in Fig. 5. The yield surface represents the boundary beyond which the material begins to yield, while Mohr's circle illustrates the stress state at a given point.

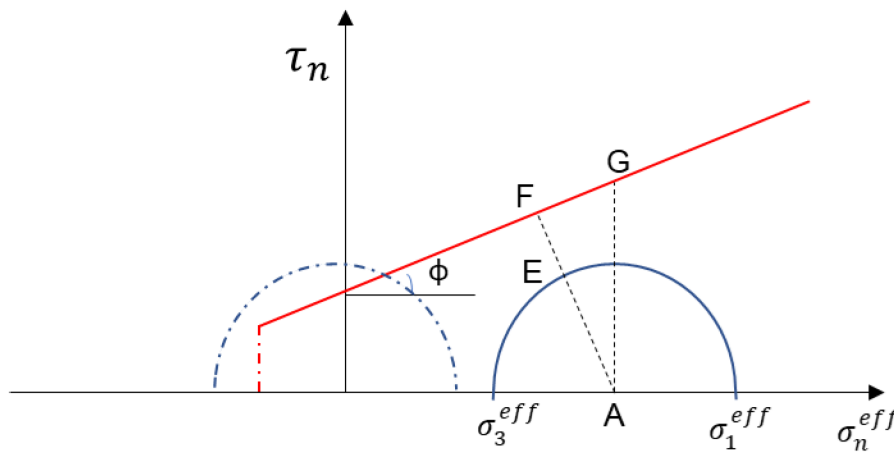


Figure 5—Material failure occurs when circle reaches yield surface.

Mathematically, SF is expressed as:

$$\text{Safety Factor (SF)} = \min \left(\frac{AF}{AE} \right) \quad (4)$$

Where $AE = \frac{1}{2}(\sigma_1^{eff} - \sigma_3^{eff})$. When safety factor greater than 1 ($SF > 1$), the fault remains stable. Conversely, as SF approaches 1, the fault is nearing potential reactivation, indicated by Mohr's circle reaching the yield surface.

Additionally, when the minimum effective principal stress σ_3^{eff} becomes tensile (negative), it indicates entry into the tensile zone. As σ_3^{eff} becomes more negative (moving further into the tensile region), the likelihood of tensile failure or hydraulic fracturing increases. This highlights the importance of considering different failure mechanisms under varying stress conditions. In general, this analysis provides a clear and structured approach to assessing fault stability.

Numerical Model Setup

Some simplifications were adopted in the model, such as neglecting capillary pressures, water salinity, thermal effects and geochemical interactions, to focus on the geomechanical response of the system due to pressure variations induced by CO₂ injection. This approach allows for a clearer understanding of the primary mechanical behaviors critical to reservoir stability.

Both one-way and two-way coupling simulations were performed in this study. In the two-way coupling approach, the permeability update is based on a permeability correlation, defined as a permeability multiplier function of the volumetric strain. For simplicity, a piecewise linear relation is assumed. The correlation can be expressed as,

$$PERMMUL = \begin{cases} 1 - 1000\epsilon_v & \text{if } -0.0005 < \epsilon_v < 0.0005 \\ 0.5 & \text{if } \epsilon_v \geq 0.0005 \\ 1.5 & \text{if } \epsilon_v \leq -0.0005 \end{cases} \quad (5)$$

Where ϵ_v is the volumetric total strain and $PERMMUL$ is the initial permeability multiplier. When the rock experiences no deformation, the volumetric strain is zero, and thus the permeability multiplier is 1. As the rock expands, the volumetric strain becomes negative (following the sign convention where compression is positive), leading to an increase in permeability. Conversely, when the rock shrinks, the volumetric strain becomes positive, resulting in a decrease in permeability. In this study, to emphasize the fault effects within the two-way coupled simulation, permeability updates for intact rock are intentionally ignored.

SPE1 Conceptual Model

As a simplified base case, we utilize a coupled reservoir-geomechanics model of the SPE1 example, introduced in the previous section, to investigate the effects of CO₂ injection. The model is modified to include a CO₂ injection well at the corner, as shown in Fig. 6 and a compositional fluid model is employed to accurately capture the complex phase behavior of CO₂, as the black oil model is not suitable for this scenario. This basic model provides a foundation for evaluating fault reactivation during CO₂ injection and serves as a reference point for subsequent simulations.

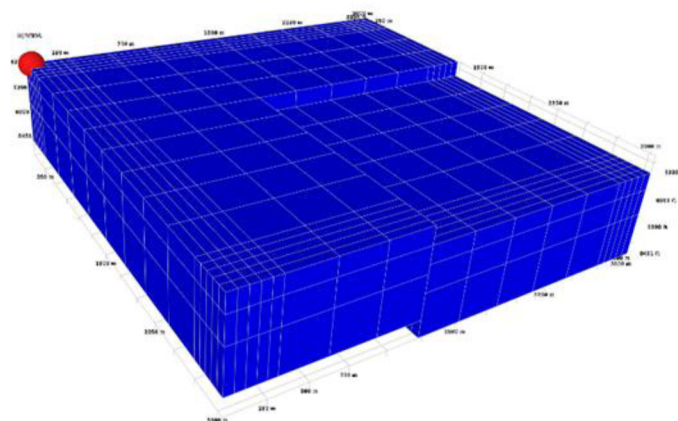


Figure 6—SPE1 model with 1 injector and 2 faults

The model specifications are as follows:

- The model consists of 972 cells, with dimensions of 1000 ft × 1000 ft × 20 ft (length × width × height) for most cells, except for those at the corners, which are 200 ft × 200 ft × 20 ft.
- No burden is included in the model.
- The rock mechanical properties are:
 - Young's modulus: 2.9e6 psi
 - Poisson's ratio: 0.3
 - Biot coefficient: 1.0
 - Bulk density: 156.07 lb./ft³
- The failure curve is defined by:
 - Cohesion: 435 psi
 - Friction angle: 30°
- The stress boundary condition is applied to the reservoir sides, with parameters:
 - C_h : 0.7
 - C_H : 0.8
 - Angle: 30°
- The fault stiffness values are:
 - Normal stiffness: 1.7683e3 psi/ft
 - Shear stiffness: 0.66e3 psi/ft
- The dip angle pairs for the faults are:
 - "X" fault: (90°, 90°)
 - "Y" fault: (90°, 0°)

The system is assumed to be isothermal and initially saturated with brine. The model neglects capillary pressure, with an irreducible water saturation (S_{wr}) of 5% and a critical gas saturation (S_{gc}) of 10%. CO₂ is injected at a constant rate of 10,000 mscf/day into a homogeneous layer with a horizontal permeability of 500 mD for a period of 10 years. For the two-way permeability coupling simulation, coupling is performed once a year. The bottom hole injection pressure is limited to a maximum of 9,000 psi. The faults are supposed to be modeled as cohesionless, with the Mohr-Coulomb failure envelope passing through the origin. To assess safety, both cohesive and cohesionless scenarios are considered, representing optimal and conservative cases, respectively.

The stress boundary condition is applied to the sides of the reservoir using regional stress, which is characterized by three components: σ_h , σ_H , and angle. The shear stress out of the horizontal plane is assumed to be zero. Specifically, σ_h and σ_H represent the minimum and maximum horizontal principal stresses, respectively. The ratios of these stresses to the vertical stress are defined as $C_h = \sigma_h / \sigma_v$ and $C_H = \sigma_H / \sigma_v$, where σ_v is the vertical stress. The angle is defined as the direction of σ_H , measured counterclockwise from the horizontal axis. Whenever possible, it is highly recommended to use measured in-situ stresses for the target field, as this can provide a more accurate representation of the stress state.

For the 1-way coupling simulation, the pressure data from the reservoir model is used to drive the geomechanical model simulation. The model boundaries are defined as follows: roller boundary conditions are applied to the sides of the model and at the bottom, while the ground surface is allowed to move freely, enabling the estimation of its displacement caused by CO₂ injection.

Large Synthetic Model

The geomechanical model, as described in the methodology section, utilizes the ‘equivalent material’ method to simulate the behavior of a large-scale reservoir with complex fault networks. The three-dimensional synthetic sector model has a domain size of 27.5 km × 27.5 km × 0.6 km (length × width × height) and is discretized into 15,000,000 grid blocks. Eleven fault structures were incorporated into the model, some of which intersect with each other, and are highlighted in purple in Fig. 7 and Fig. 8. The reservoir is assumed to be bounded by a low-permeability caprock at the top and an under burden at the bottom, although the caprock is not explicitly included in the simulation for this study, making the reservoir a suitable target storage aquifer for CO₂ sequestration.

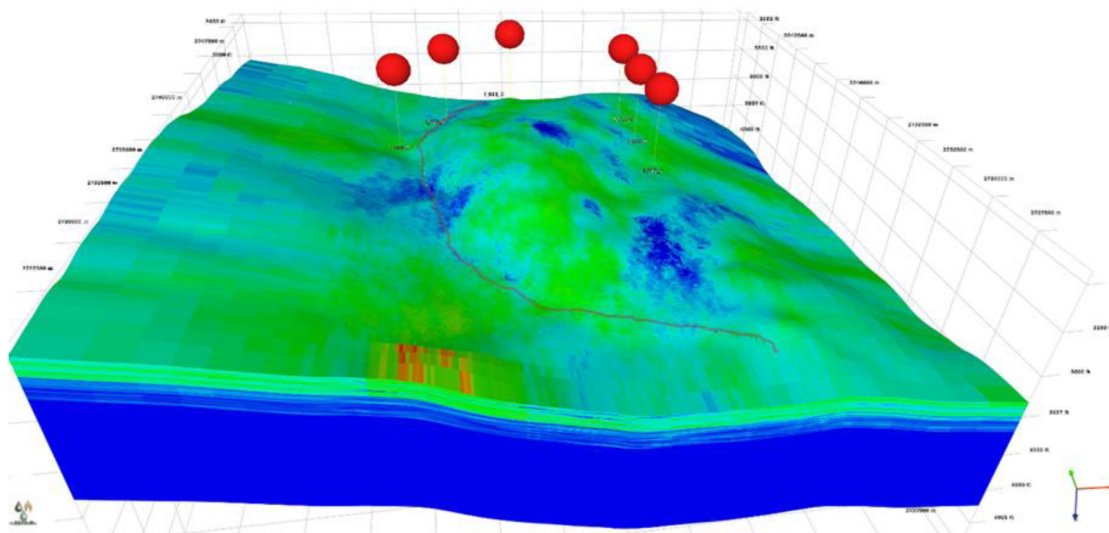


Figure 7—3D Synthetic Sector Model with Highlighted Faults

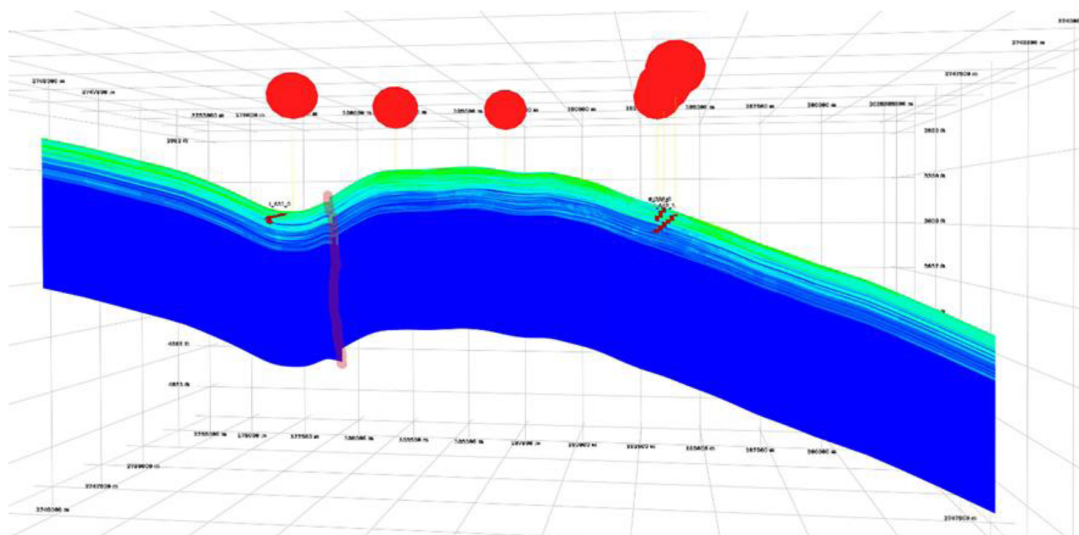


Figure 8—Cross-section Synthetic Sector Model with Highlighted Faults

The model specifications are as follows:

- The model consists of 15 million cells, with a grid size of $3270 \times 1933 \times 1000$.
- The model does not include an overburden. However, due to the properties and size of the model, the side burden and underburden can be considered as implicitly included.
- A linear poro-elastic model is employed to simulate the behavior of the reservoir.
- The rock mechanical properties are:
 - Young's modulus: 2.9e6 psi
 - Poisson's ratio: 0.3
 - Biot coefficient: 1.0
 - Bulk density: 156.07 lb./ft³
- The failure curve is defined by:
 - Cohesion: 435 psi
 - Friction angle: 30°
- The stress boundary condition is applied to the reservoir sides, with parameters:
 - C_h : 0.7
 - C_H : 0.8
 - Angle: 30°
- The fault stiffness values are:
 - Normal stiffness: 1.7683e3 psi/ft
 - Shear stiffness: 0.66e3 psi/ft

The fluid properties used in this simulation are identical to those employed in the previous case. As illustrated in Fig. 7, the model includes 6 CO₂ injection wells, represented by red bubbles, which are grouped into two categories based on their location and injectivity. One group of wells is situated along the faults in a region of relatively high permeability roughly from 12 mD, resulting in high injectivity. The other group is located farther away from the faults in an area of lower permeability roughly around 0.12 mD, resulting in lower injectivity. It is worth noting that, from a safety perspective, injectors and producers are not suggested to be placed in close proximity to faults or fractures. However, the configuration used in this study is intentionally designed to simulate worst-case scenarios, where injectors are placed near faults or fractures to examine the potential consequences.

CO₂ is injected at a constant rate for both groups of wells over various time periods, ranging from 25 to 80 years, until the maximum bottom hole injection pressure of 10,000 psi is reached. For the two-way permeability coupling simulation, coupling is performed every 6 months.

The injection rates for the two groups of wells are as follows:

- Wells located close to faults (high injectivity):
 - I₁₁: 1,000 mscf/day
 - I₂₁: 10,000 mscf/day

- I_31: 10,000 mscf/day
- Wells located close to tight formations (low injectivity):
 - I_10: 5,000 mscf/day
 - I_22: 5,000 mscf/day
 - I_42: 4,000 mscf/day

By varying the injection rates, selecting wells in regions with high or low injectivity, and considering different distances to faults, a targeted sensitivity analysis is conducted to investigate the impact of these parameters on reservoir integrity during CO₂ injection. This analysis focuses on the individual and combined effects of:

- Injection rates: varying the rates at which CO₂ is injected into the reservoir (e.g., 1,000 mscf/day, 5,000 mscf/day, 10,000 mscf/day)
- Injectivity: considering the differences in injectivity between wells located in regions with high injectivity (close to faults) and those located in regions with low injectivity (far from faults)
- Distance to faults: examining the effects of injecting CO₂ at varying distances from faults, including both near-fault and far-fault scenarios

This limited analysis provides insights into the key impacts of these parameters on the safety and efficacy of CO₂ storage, striking a balance between complexity and computational efficiency.

Results and Discussion

Permeability updates and CO₂ injection volume

To highlight the impact of stress-sensitive faults on permeability, the fault stiffness was set to a relatively low value. This allows for a more pronounced demonstration of how permeability can change significantly during injection when faults are sensitive to stress.

The results of the two-way coupling simulation are compared to the one-way coupling simulation in Fig. 9 for the SPE1 case. The differences between the two simulations are clearly visible and highlighted in red, illustrating the significant impact of permeability updates on the simulation results.

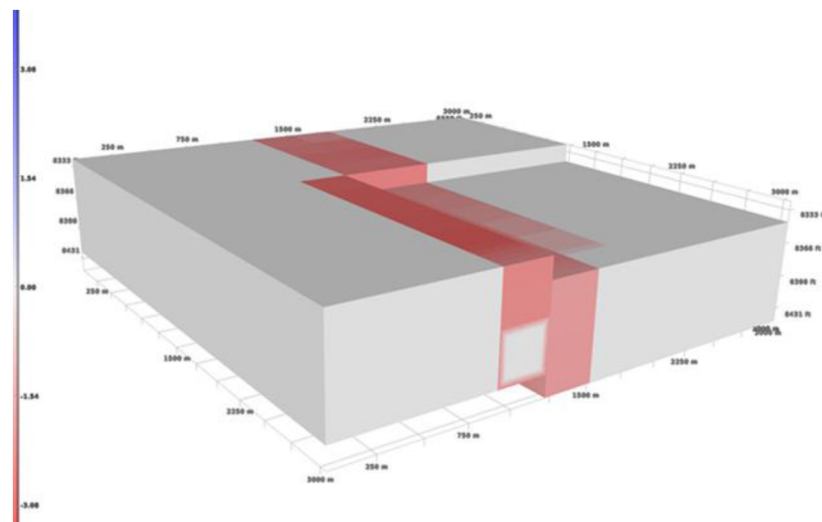


Figure 9—comparison of simulation results from one-way and two-way coupling for permeability after 6 years injection.

Fig. 10 shows the cumulative volumes of CO₂ injected over a ten-year injection period. Notably, the two-way simulation reaches the injection bottom-hole pressure (BHP) constraint earlier than the one-way simulation, resulting in a lower total volume of CO₂ injected. This demonstrates that accurately accounting for the coupling between flow and mechanics is crucial for assessing the safe storage capacity of a reservoir, as it can significantly impact the estimated volume of CO₂ that can be injected.

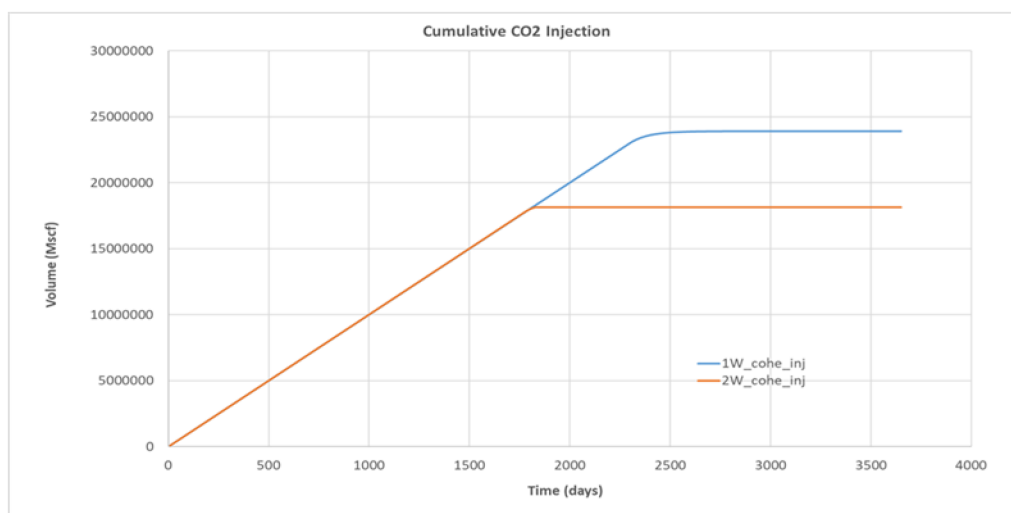


Figure 10—Comparison of cumulative CO₂ injection volumes between one-way and two-way coupling

Fig. 11 illustrates the impact of geomechanics on well bottom-hole pressure (BHP) for the large synthetic case, using two-way coupling, with scenarios considering the presence and absence of cohesion, resulting in a substantially shortened injection period and a significant reduction in injection volume.

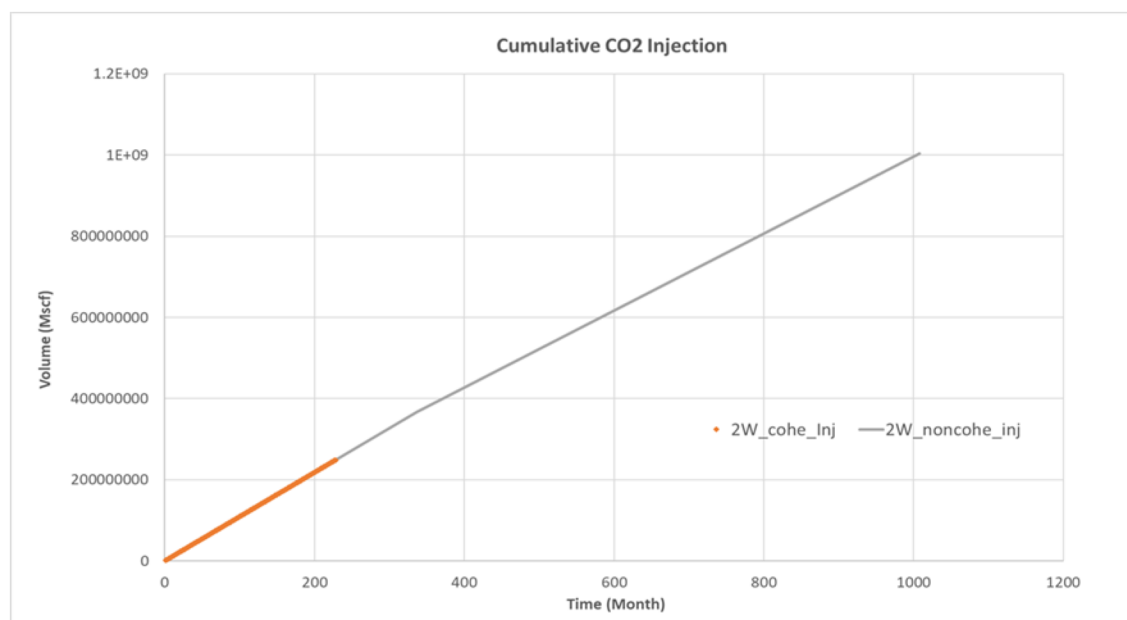


Figure 11—Comparison of cumulative CO₂ injection volumes between cohesive and cohesionless fault scenarios.

Evolution of the displacement

Quantifying deformation caused by CO₂ injection is crucial, as it directly impacts well and surface integrity, with even small changes potentially posing significant risks. This deformation can be monitored at any point in the subsurface or at ground level by calculating the vertical displacement resulting from compaction or inflation phenomena caused by CO₂ injection. Fig. 12 (a) illustrates the vertical displacement for the 1-way coupling SPE1 case, which exhibits the most significant displacement after 10 years of injection. The reservoir deformation, characterized by dilation in this case, leads to a maximum ground-level uplift of 0.17 ft. The largest displacements are observed near the injector. It is unlikely that this level of reservoir and surface uplift will result in ground surface failure over time. The evolution of vertical displacement in a representative cell close to the fault is shown in Fig. 12 (b). As CO₂ is injected into the reservoir, the formation begins to expand, and dilation is observed during the injection period.

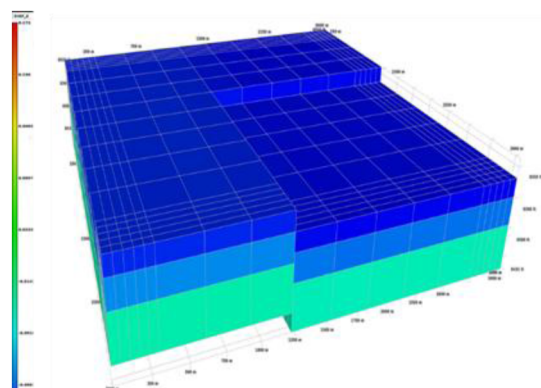


Figure 12-(a)—1-way SPE1 vertical displacement

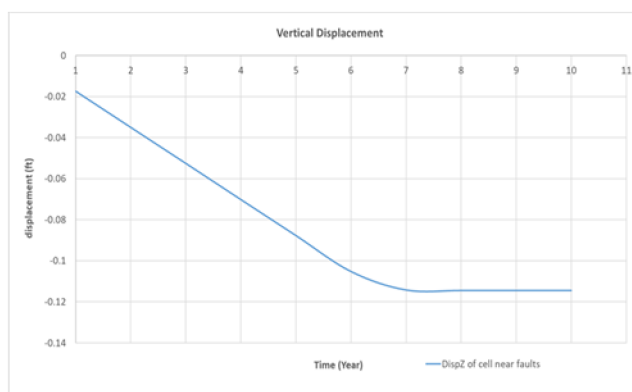


Figure 12-(b)—near-fault cell vertical displacement

Fig. 13 illustrates the vertical displacement for the 2-way coupling cohesionless case, considered the worst scenario, for the synthetic sector model. After 85 years of injection, the maximum displacements occur near the injection wells, reaching a maximum value of 0.63 ft.

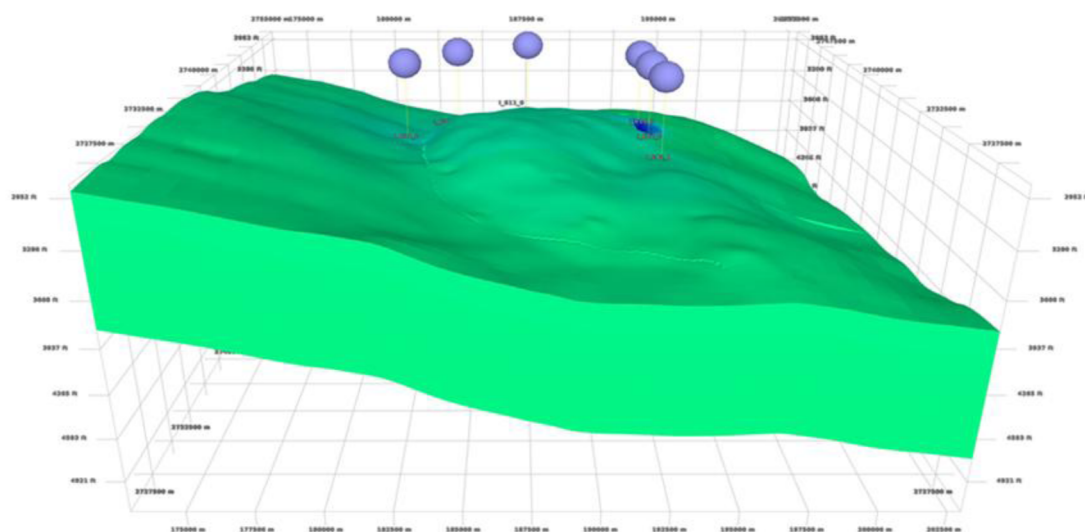


Figure 13—The vertical displacement for Synthetic Sector model

Fig. 14 and 15 show the faults' normal displacement (or "opening") and shear displacement (or "sliding") after 20 years of CO₂ injection, respectively. The field-scale case study reveals complex spatial patterns in the reservoir's response, with notable multi-fault interactions observed at the fault intersections.

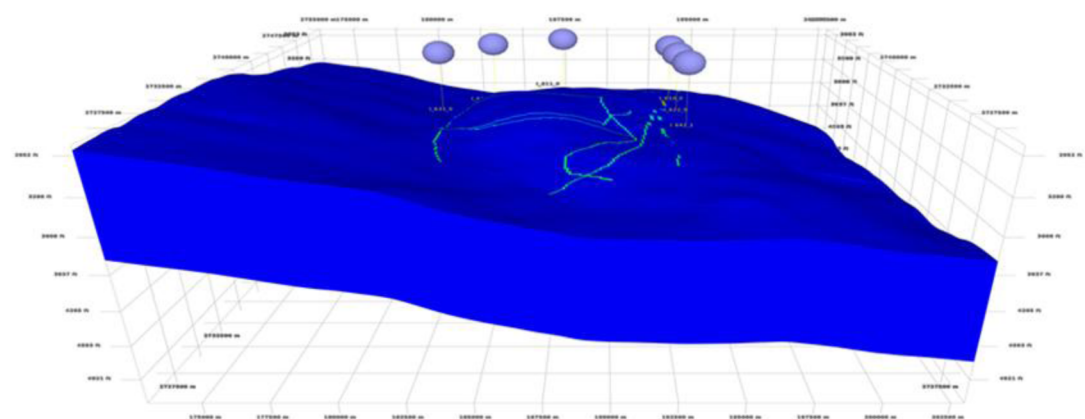


Figure 14—The faults normal displacement for Synthetic Sector model

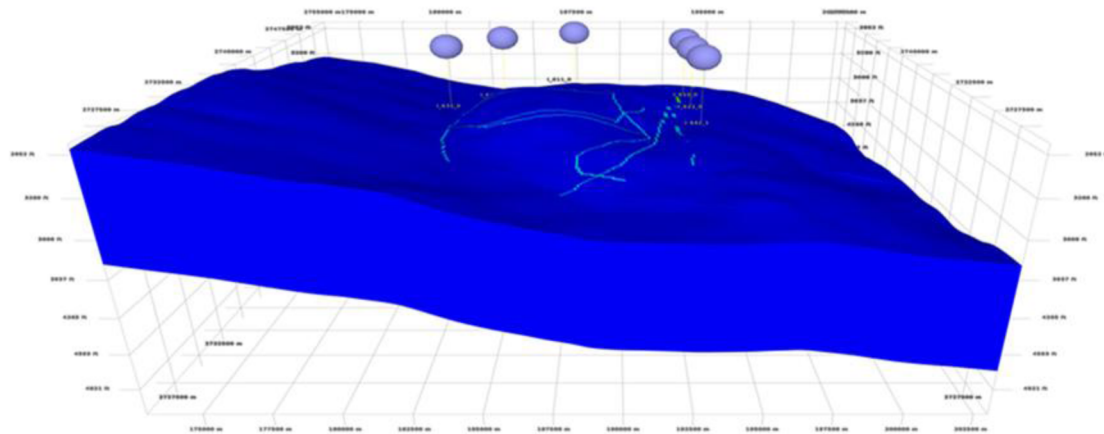


Figure 15—The faults shear displacement for Synthetic Sector model

The stress field is illustrated in Fig. 16 (a)-(f), which shows the response of the reservoir to CO₂ injection. The near-well regions exhibit an immediate response to injection, whereas the far-field regions display delayed responses, with pressure and stress continuing to evolve over time. The horizontal stress components (σ_{xx} , σ_{yy}) are significantly affected by the presence of faults, as seen in the figure. In contrast, the other stress components are less influenced by the faults, which can be attributed to the fact that most of the faults penetrate the reservoir nearly vertically.

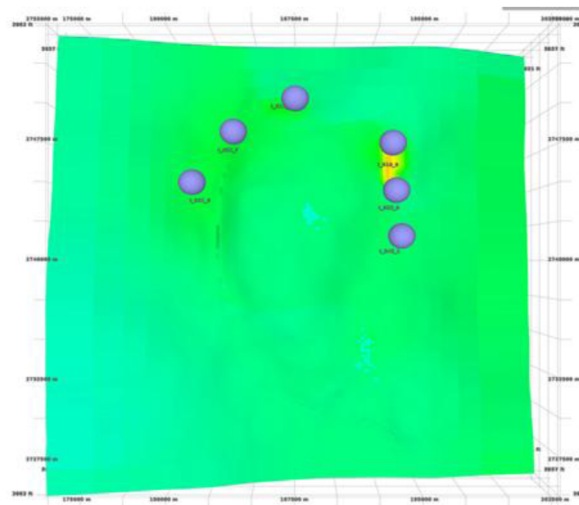


Figure 16-(a)—top view of σ_{xx} field

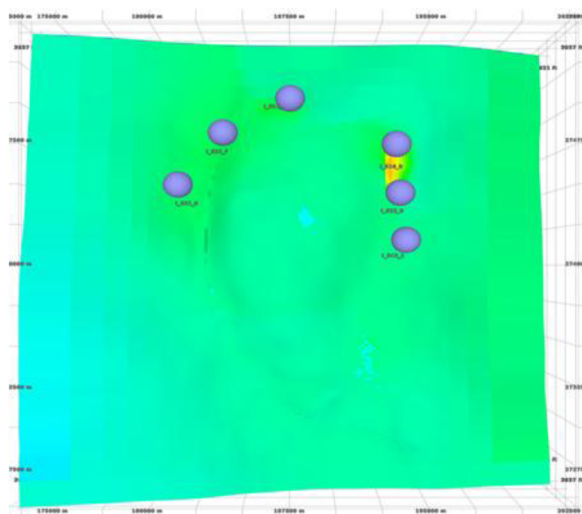


Figure 16-(b)—top view of σ_{yy} field

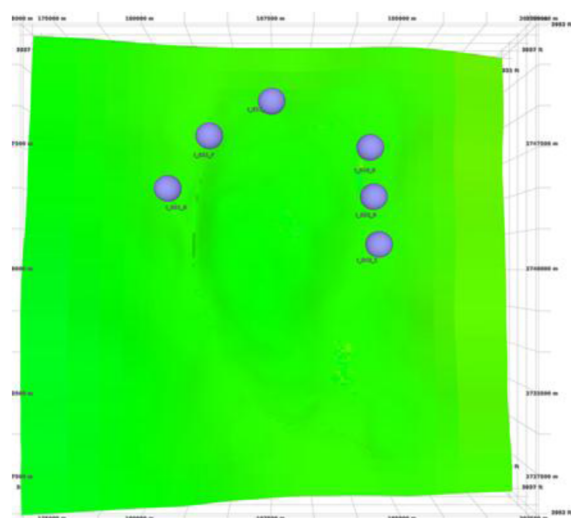


Figure 16-(c)—top view of σ_{zz} field

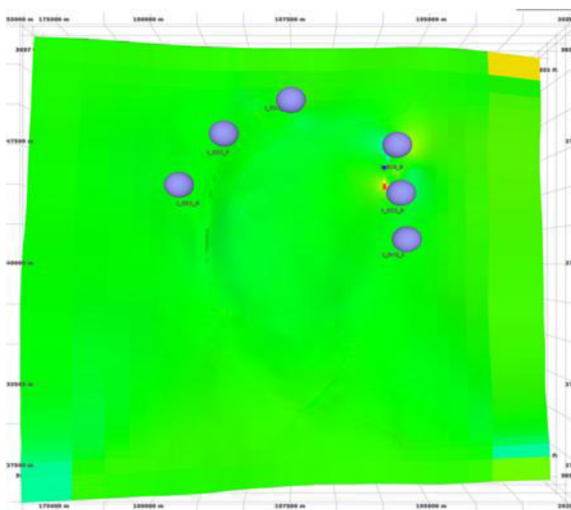


Figure 16-(d)—top view of σ_{xy} field

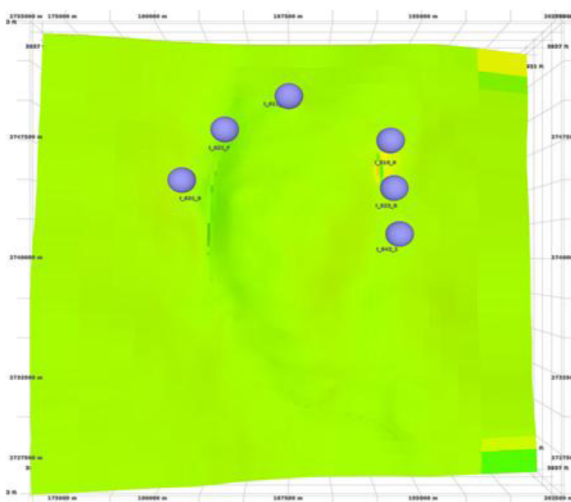


Figure 16-(e)—top view of σ_{xz} field

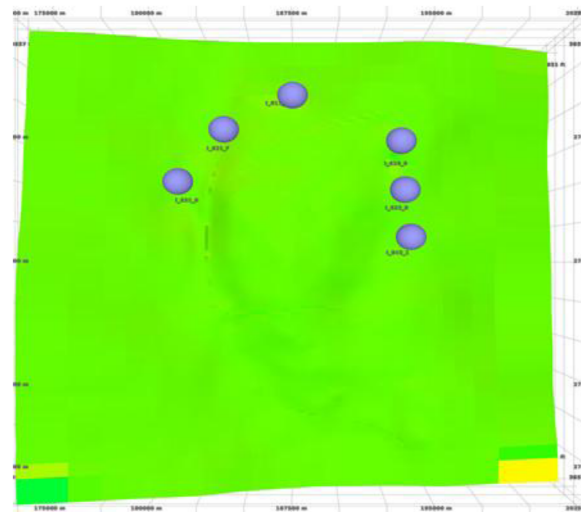


Figure 16-(f)—top view of σ_{yz} field

Fault Integrity

Although all faults represent mechanical discontinuities that can be activated during CO₂ injection, the severity of their potential impact on storage security varies significantly depending on their vertical extent and interaction with sealing formations. The presence of an intact caprock and overburden above these faults provides additional barriers against potential CO₂ migration, effectively reducing the risk associated with these faults compared to those that have broken through to the surface. In the absence of overburden information, our cases can be considered as representative of lower-bound scenarios, where faults with inherent mechanical weakness are more likely to be activated. Building on this understanding, the stability of faults is influenced by two key factors: the total stress around the fault and the fluctuation of boundary conditions. To evaluate fault stability, we apply a safety factor established by the Mohr-Coulomb destabilization failure criterion, which serves as a screening tool to assess the potential for fault activation. To examine the impact of fault stability on CO₂ storage, we consider an extreme scenario for both the SPE1 case and the synthetic case, where injection wells are placed in close proximity to the faults. This scenario allows us to investigate the potential risks associated with fault activation and CO₂ migration, and the results of this analysis are presented in the following figures.

The safety factors for cells near the fault and near the injectors in the SPE1 case are presented in Figure 17, which compares the results for two-way coupling with and without cohesion. Fig. 17(a) shows the evolution of the safety factor over the injection period, while Fig. 17(b) provides a detailed map of the safety factor differences across the model cells at the 6th year of injection. The results indicate that by the end of the 5th year of injection, the safety factor for both cases falls below 1, indicating that the reservoir integrity is compromised thereafter. Notably, the cohesionless case reaches the failure envelope approximately half a year earlier than the cohesion case, suggesting that the absence of cohesion accelerates the loss of reservoir integrity.

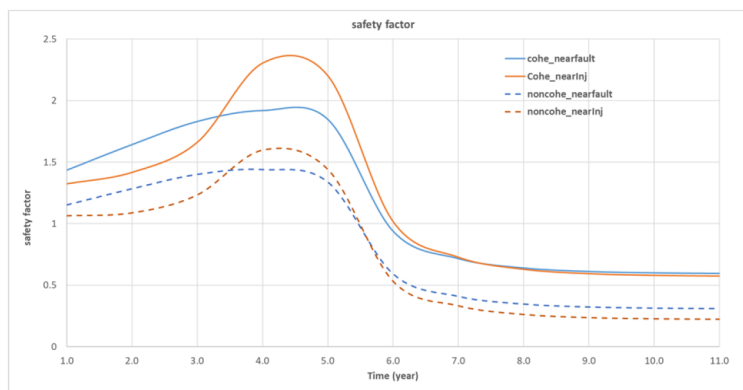


Figure 17-(a)—Safety Factor of selected cells

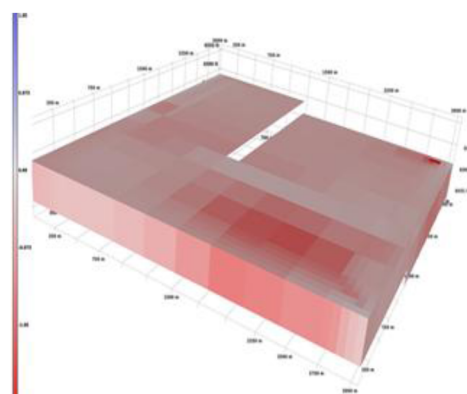


Figure 17-(b)—Diff for safety factor

Figure 18 presents a comparative histogram of the safety factors for both cases at the end of the injection period. As shown in Fig. 18(a), the majority of cells in the cohesive case have a safety factor above 1.4, indicating that they remain intact. In contrast, Fig. 18(b) reveals that the majority of cells in the cohesionless case have a safety factor below 0.3, suggesting a significant loss of integrity.

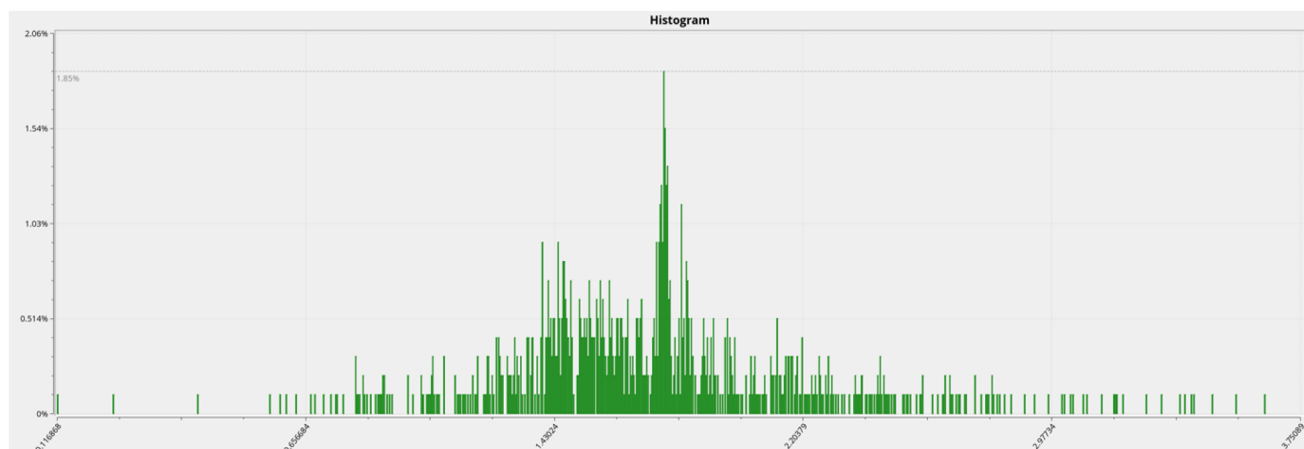


Figure 18-(a)—Safety Factor distribution of SPE1 cohesion case

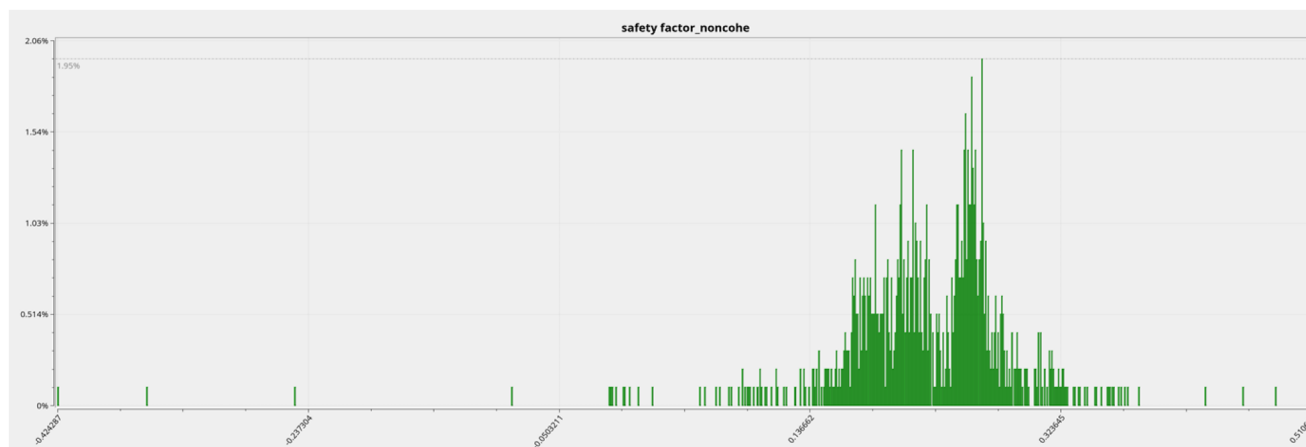


Figure 18(b)—Safety Factor distribution of SPE1 cohesionless case

To identify the development of instability in the synthetic sector model, a filter with a safety factor value less than 1 was applied as a screening tool for the two-way coupling simulations with and without cohesion. The evolution of the safety factor over the injection period is illustrated in Figure 19 (at the 5th year) and Figure 20 (at the 20th year). Specifically, Fig. 19(a) presents the front and top views of the cohesion case, while Fig. 19(b) shows the corresponding views for the cohesionless case. The results indicate that failure initiates in the low-permeability or low-injectivity regions for both cases. However, a notable difference is observed between the two cases: in the cohesionless case, significant failure regions are evident near the faults, whereas the cohesion case remains intact in these areas. As injection continued for an additional 15 years, the failure region continued to evolve and expand in the low-injectivity area. Notably, the regions close to the faults exhibited distinct behavior between the two cases. For the cohesion case, as shown in Fig. 20(a), failure was only observed in the vicinity of the injector with the highest injection rate. In contrast, the cohesionless case, as illustrated in Fig. 20(b), developed an extensive failure region surrounding the faults, indicating a more pronounced and widespread loss of stability.

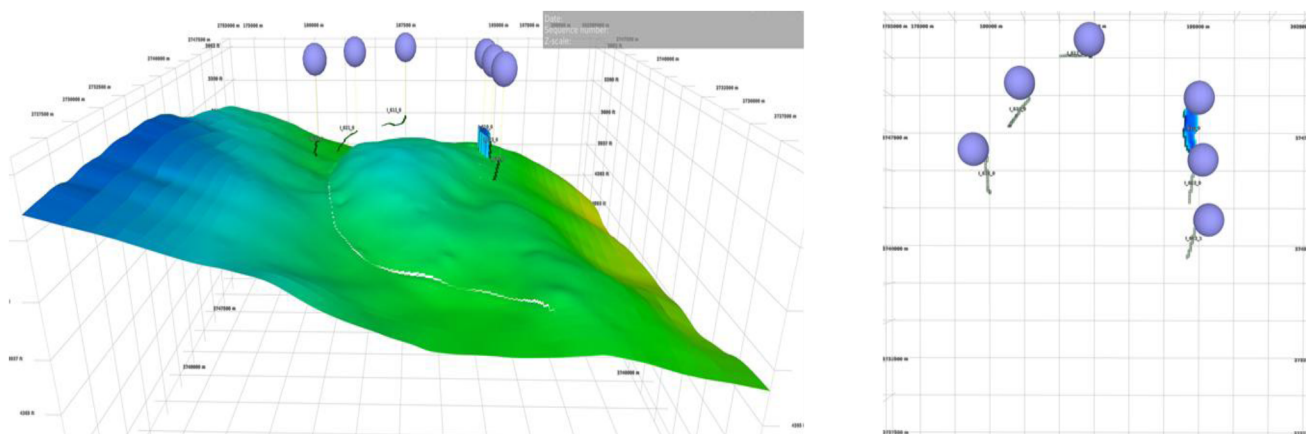


Figure 19(a)—Regions of instability for the cohesion case at the 5th year of injection ($SF < 1$)

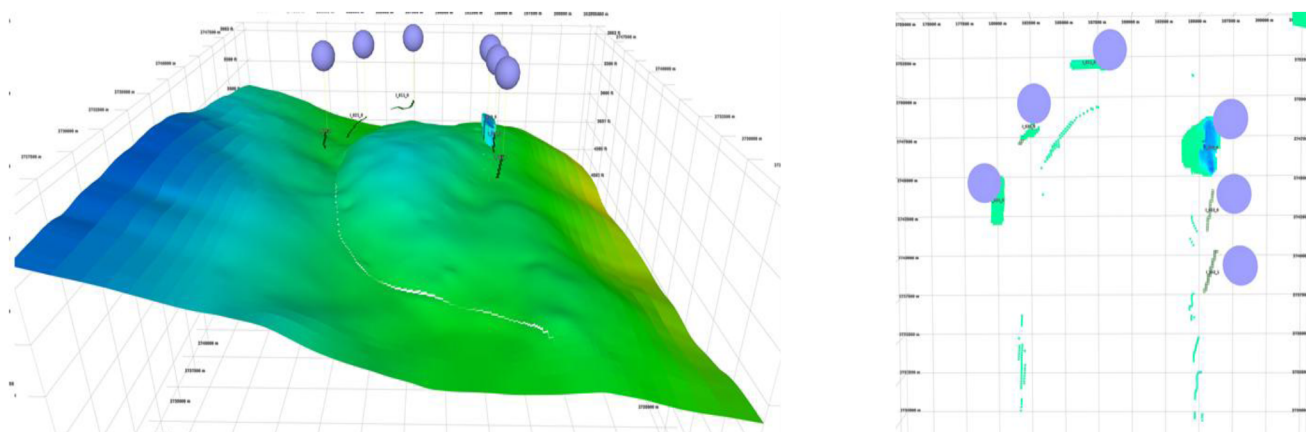


Figure 19-(b)—Regions of instability for the cohesionless case at the 5th year of injection ($SF < 1$)

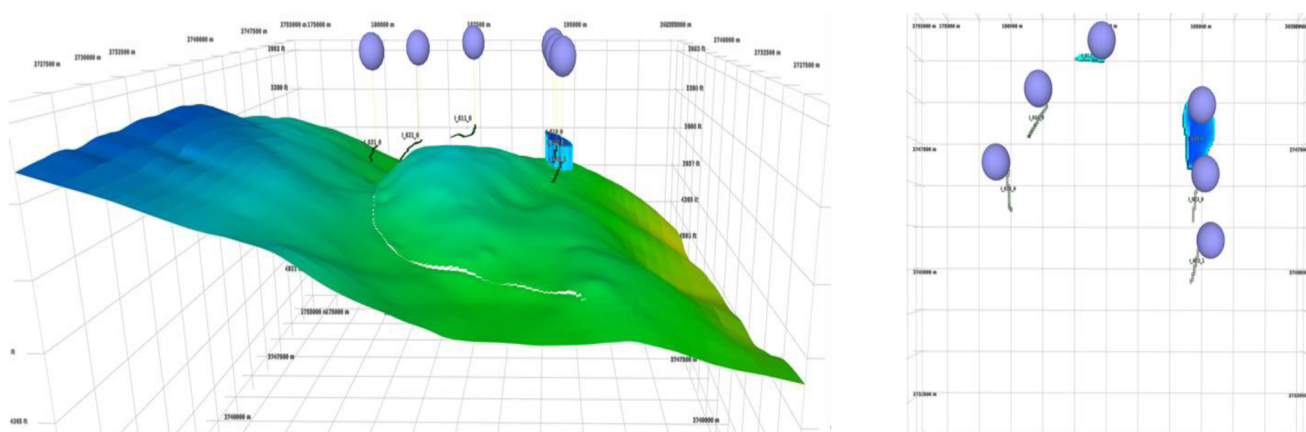


Figure 20-(a)—Regions of instability for the cohesion case at the 20th year of injection ($SF < 1$)

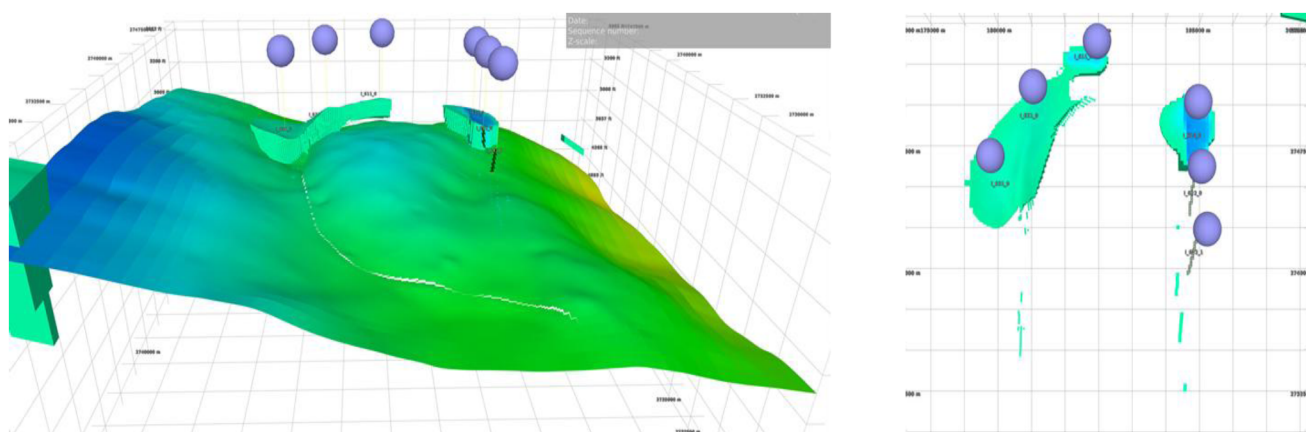


Figure 20-(b)—Regions of instability for the cohesionless case at the 20th year of injection ($SF < 1$)

Figure 21 presents a comparative histogram of the safety factors for both cases at the end of the 20th year of injection. The results show a stark contrast between the two cases. As illustrated in Fig. 21(a), the cohesive case exhibits a majority of cells, approximately 97%, with a safety factor above 1.5, indicating that they remain stable and intact. In contrast, Fig. 21(b) reveals that around 10% of cells in the cohesionless case have a safety factor below 1.0, suggesting a moderate loss of integrity.

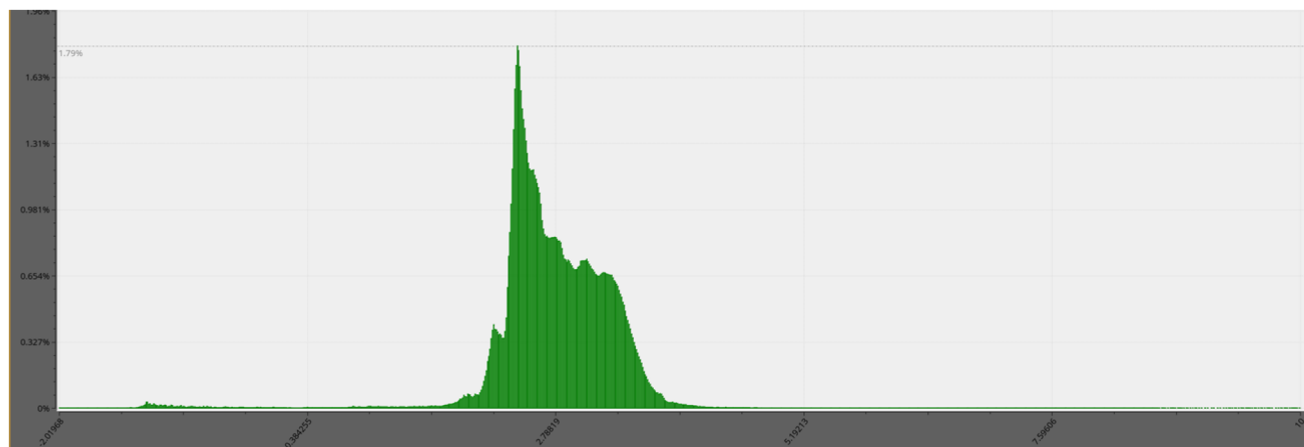


Figure 21-(a)—Safety Factor distribution of for Synthetic Sector model cohesion case

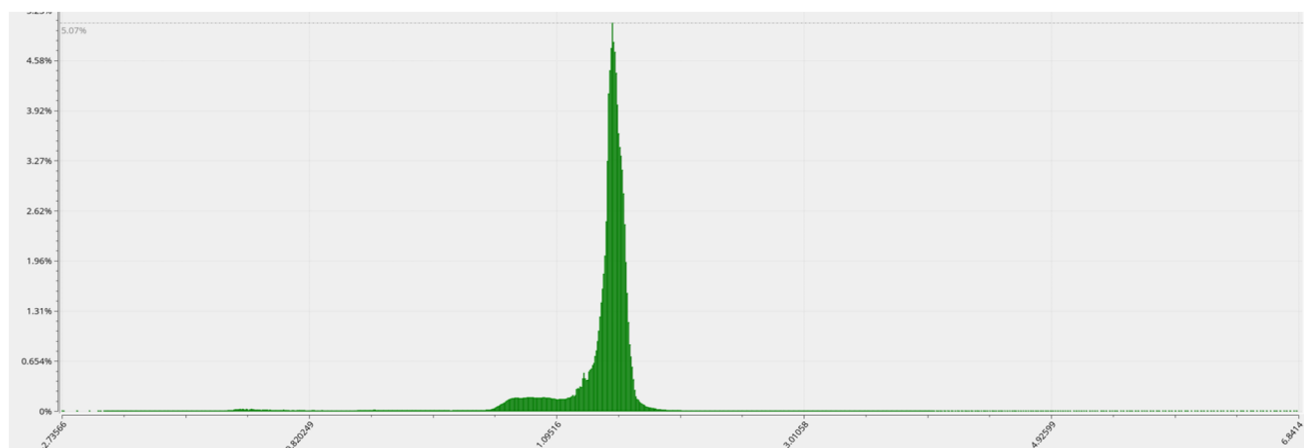


Figure 21-(b)—Safety Factor distribution of for Synthetic Sector model cohesionless case

The simulation results reveal that rock mechanical properties and reservoir heterogeneities play a crucial role in determining fault stability, as evidenced by a clear correlation between the failure indicator and these parameters. The stability of the reservoir, including the faults, is influenced by a complex interplay of factors, including CO₂ injection rates, fault characteristics (location, stiffness), and the combination of rock formations (matrix permeability) in the system. Two primary mechanisms of reservoir instability induced by CO₂ injection are identified:

- When a fault is proximal to or intersects with an injector, CO₂ injection into the reservoir causes an increase in pore pressure within the fault, leading to differential total stress changes around the fault that compromise its stability.
- Low formation permeability hinders fluid flow, leading to pore pressure buildup, and the SF indicator reveals that tensile failure dominates around injectors, with shear failure occurring to a lesser extent.

The aquifer matrix permeability is a critical property, as higher permeabilities facilitate easier fluid movement and significantly reduce pressure buildup at the injector. These findings underscore the importance of carefully selecting the location of injector wells to balance high injection volumes with the need to avoid formation and fault failure.

Conclusions

This research evaluates geomechanical risks associated with CO₂ injection, focusing on the impact of well placement, injection rates, and formation injectivity. Using both a simplified SPE1 model and a large synthetic sector model case, we conducted a comprehensive numerical simulation investigation. The analysis highlights how well placement, injection rates, and formation injectivity influence geomechanical risks such as fault reactivation, zone of failure, and surface uplifting during CO₂ injection. We examined several scenarios and parameters to understand the influence of uncertainties in geo-mechanical and geological factors.

The findings provide valuable insights for assessing the safety and efficiency of CO₂ storage, emphasizing the importance of accurately predicting and managing stress changes during CO₂ injection to ensure sustainable energy management and environmental protection. Key findings include:

- The comparison of one-way and two-way coupling strategies, which demonstrated the superiority of the two-way coupled approach in capturing the onset of instability at the fault.
- The analysis of fault behavior due to CO₂ injection, including the impact of permeability update, fault stress behavior, and fault displacement.
- The application of the Safety Factor (SF) indicator to evaluate reservoir integrity.

The findings underscore the importance of geomechanical properties like fault stiffness and cohesion in maintaining reservoir integrity. Furthermore, detailed knowledge of pre-existing faults, reservoir heterogeneities, and geomechanical properties is crucial for a comprehensive understanding of the reservoir's behavior under injection conditions.

Future research will build on these findings by utilizing the advanced elasto-plastic nonlinear feature of our in-house geomechanics simulator. This will enable more extensive uncertainty analyses, enhancing risk assessment and helping to better mitigate fault reactivation during CO₂ injection operations.

Acknowledgement

The authors would like to thank Lei Qiao and Meng Shen of the Strategic Modeling Team in Boston for their contributions to the Geomechanics simulator. We also extend our gratitude to the Strategic Modeling Team in Dhahran for their support.

Reference

- Ali, M., Jha, N.K., Pal, N., Keshavarz, A., Hoteit, H., Sarmadivaleh, M., 2022. Recent advances in carbon dioxide geological storage, experimental procedures, influencing parameters, and future outlook. *Earth Sci. Rev.* **225**, 103895.
- Ji-Quan Shi, Wenzhuo Cao, Sevet Durucan, Anna Korrea. Assessment of CO₂ injection induced fault reactivation and reservoir performance conformance at the In Salah CO₂ storage site. 2024. 17th International Conference on Greenhouse Gas Control Technologies GHGT-17.
- Andrea Ferrari, Stefano Nardean, Silvia Monaco, San Donato Milanese, Timur Garipov, Bel Air, Stefano Mantica, Alberto Cominelli, and San Donato Milanese. Effects of Flow and Geomechanics Coupling in Faulted Reservoirs for CO₂ Storage. 2024. SPE-220000-MS
- A. Gillioz, Yverdon-les-Bains, F. Verga and C. Deangeli. Critical Parameters for Caprock Tensile Failure Induced by CO₂ Injection into Aquifers. 2024. SPE-220097-MS
- Evgenii Kanin, Igor Garagash, Sergei Boronin, Svetlana Zhigulskiy, Artem Penigin, Andrey Afanasyev, Dmitry Garagash and Andrei Osiptsov. CO₂ storage in deep saline aquifers: evaluation of geomechanical risks using integrated modeling workflow. 2023.
- Lei Qiao, Meng Shen and Ali Dogru. Coupled geomechanics and reservoir simulation for gigantic naturally fractured reservoirs. 2024. IPTC-23586-MS.
- Zoback, M. D. (2007). Reservoir Geomechanics. Cambridge University Press. <https://doi.org/DOI:10.1017/CBO9780511586477>
- Ting Xiao, Ting Chen, Zhiwei Ma, Hailong Tian, Saro Meguerdijian, Bailian Chen, Rajesh Pawar, Lianjie Huang, Tianfu Xu, Martha Cather, Brian McPherson. A review of risk and uncertainty assessment for geologic carbon storage. *Renewable and Sustainable Energy Reviews*. Volume **189**, Part B, January 2024, 113945

G.N. Pande, G. Beer, J.R. Williams, Numerical Methods in Rock Mechanics, John Wiley & Sons Inc, Chichester; New York, 1990.

Schlumberger, VISAGE Technical Description, Version 2012, (2012).